Software Requirements Specification

for

(Communication Module)

National Freelance and Skill Verification Platform

Version 1.0 Approved

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Revision History

Name	Date	Reason For Changes	Version
Nisha Zafran	28-03-2026	Initial Draft	1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a comprehensive and detailed description of the Communication Module within the National Freelance and Skill Verification Platform. This module serves as a critical subsystem that enables seamless, secure, and efficient interaction between freelancers, clients, and administrative users.

The Communication Module is designed to support various forms of communication including real-time messaging, Media sharing, notifications, and Meeting links. Its primary goal is to eliminate communication barriers that often exist in freelance ecosystems by providing a centralized, reliable, and transparent communication channel.

This document defines the functional and non-functional requirements of the module, outlines system behavior under different conditions, and establishes a clear understanding between stakeholders, developers, and testers. It ensures that all parties involved have a unified vision of how the communication system should operate within the broader platform

1.2 Document Conventions

This document follows standard SRS conventions based on IEEE guidelines. All requirements are uniquely identified using labels such as CM-FR-01, CM-FR-2, etc., to ensure traceability and clarity. The keyword “shall” is used to indicate mandatory requirements, while “may” indicates optional functionality.

Priority levels (High, Medium, Low) are assigned to features based on their importance to system functionality. Any incomplete or pending information is marked as TBD (To Be Determined)**.**

1.3 Intended Audience and Reading Suggestions

This document is intended for multiple stakeholders involved in the project. Developers will use it as a foundation for implementation, testers will rely on it to design test cases, and project managers will use it for planning and monitoring progress. Additionally, UI/UX designers will refer to interface-related sections, while stakeholders and evaluators will use it to understand the system’s scope and capabilities.

Readers are encouraged to begin with the Introduction and Overall Description sections to understand the system context, followed by the System Features section for detailed functional requirements. Non-functional requirements should be reviewed to understand performance, security, and quality expectations.

1.4 Product Scope

The Communication Module is a core component of the National Freelance and Skill Verification Platform, designed to facilitate structured and secure interaction among users. In freelance environments, effective communication is essential for project success, requirement clarity, and conflict avoidance.

This module enables users to exchange messages in real time, share files, image and videos related to projects, receive notifications, and conduct meetings through integrated third-party platforms. It directly contributes to the platform's broader objectives of building trust, improving collaboration, and ensuring transparency in freelance engagements.

By providing a centralized communication infrastructure, the module reduces dependency on external tools and ensures that all interactions are recorded, secure, and accessible for future reference.

1.5 References

- IEEE 830 Software Requirements Specification Standard
- Software Project Management Course Material
- National Freelance Platform Business Case Document
- REST API and WebSocket Communication Standards

2. Overall Description

2.1 Product Perspective

The Communication Module is part of a larger microservices-based architecture and functions as the primary interaction layer between users. It is not a standalone system but is tightly integrated with module User Management.

It acts as a bridge that connects users during various stages of project execution, from initial discussions to final delivery. The module also ensures that communication data is stored and can be referenced in cases such as disputes or performance evaluations.

2.2 Product Functions

- Real-time messaging
- Multimedia management (file, images, videos sharing)
- Generate links for online meetings (zoom or google meet)
- One to one communication and Group chats

- Message status indicators (sent/read)
- Message Storage
- Basic Communication Interface Support
- Notification of received messages
- API/Service for Messaging
- End to end encryption of messages

2.3 User Classes and Characteristics

User Class	Description
Freelancers	Frequent users, moderate technical skills
Clients	Moderate usage, business-oriented
Admins	High control, monitors communication

2.4 Operating Environment

- Web browsers (Chrome, Edge)
- Backend: Cloud-based servers
- Database: Mern
- APIs: RESTful services

2.5 Design and Implementation Constraints

- Must integrate with existing modules
- Use secure communication protocols (WebSockets)
- Limited academic development time
- Must support scalability

2.6 User Documentation

- User manual (PDF)
- In-app help section
- FAQ and tutorials

2.7 Assumptions and Dependencies

Assumptions:

- Users have internet access
- Third-party APIs (Zoom/Google Meet) are available

Dependencies:

- User module
- Notification Service
- Database Service

3. External Interface Requirements

3.1 User Interfaces

- Chat window with message threads
- Send button, attachment icon
- Notification pop-ups
- Text box for writing message
- Button to generate meeting link

3.2 Hardware Interfaces

- Smartphones, laptops, desktops
- Microphone and camera (for meetings)

3.3 Software Interfaces

- Authentication API
- Notification API
- Message Storage Service
- File Storage Service
- Third-party APIs (Zoom, Google Meet)

3.4 Communications Interfaces

- Protocols: HTTP, HTTPS, WebSockets
- Encryption: SSL/TLS
- Data format: JSON
- Real-time sync using sockets

4. System Features

4.1 Real-Time Messaging System

4.1.1 Description and Priority

Allows instant communication between users.
Priority: High

4.1.2 Stimulus/Response Sequences

- User sends message → System delivers instantly
- User receives message →Notification shown
- Network failure → Message queued and retried

4.1.3 Functional Requirements

- **CM-FR-1:** System shall allow users to send text messages.
- **CM-FR-2:** System shall deliver messages in real-time.
- **CM-FR-3:** System shall show message timestamps.
- **CM-FR-4:** System shall display delivery status (sent/delivered/read).
- **CM-FR-5:** System shall handle message failure and retry.
- **CM-FR-6:** System shall ensure security of data (encryption).

4.2 File Sharing Feature

4.2.1 Description and Priority

Allows sharing of files, images, videos within chat.
Priority: High

4.2.2 Stimulus/Response Sequences

- User uploads file, image, video → System validates → Sends
- Invalid file, image, video → Error displayed

4.2.3 Functional Requirements

- **CM-FR-7:** System shall allow Media attachments.
- **CM-FR-8:** System shall support images, PDFs, docs, videos.
- **CM-FR-9:** System shall enforce file, image, video size limits.

4.3 Voice/Video Meeting Integration

4.3.1 Description and Priority

Supports external meeting links.
Priority: Medium

4.3.2 Stimulus/Response Sequences

- User creates meeting → Link generated
- Other users clicks → Redirect to platform

4.3.3 Functional Requirements

- **CM-FR-10:** System shall generate meeting links.
- **CM-FR-11:** System shall integrate with Zoom/Google Meet.
- **CM-FR-12:** System shall store meeting link in history.

4.4 Notification Systems

4.4.1 Description and Priority

Alerts users about messages.
Priority: High

4.4.2 Stimulus/Response Sequences

- Message received → Notification Triggered

4.4.3 Functional Requirements

- **CM-FR-13:** System shall notify users of new messages.
- **CM-FR-14:** System shall support in-app notifications.

4.5 Group Chat System

4.5.1 Description and Priority

Supports team communication.
Priority: High

4.5.2 Stimulus/Response Sequences

- User creates group → Members added
- Message sent → Visible to all

4.5.3 Functional Requirements

- **CM-FR-16:** System shall allow group chat creation.
- **CM-FR-17:** System shall support multiple participants.
- **CM-FR-18:** System shall allow role-based permissions.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system must ensure that messages are delivered within two seconds under normal conditions and should support thousands of concurrent users without performance degradation.

5.2 Safety Requirements

The system must prevent data loss and ensure reliable storage of communication records. Backup mechanisms must be implemented to safeguard user data.

5.3 Security Requirements

All communication must be encrypted. The system must protect against unauthorized access, spam, and malicious content.

5.4 Software Quality Attributes

The system should be reliable, scalable, and user-friendly. It should also be maintainable and adaptable to future enhancements.

5.5 Business Rules

Only registered users can communicate, and communication is typically initiated after job engagement. Admins have the authority to monitor communication in case of disputes.

6. Other Requirements

The system should support future enhancements such as multilingual communication and advanced analytics.

7. Appendix A: Glossary

Term	Definition
API	Application Programming Interface
Real-time	Instant communication
Escrow	Secure payment holding system
WebSocket	Protocol for real-time communication

8. Appendix B: Analysis Models

- Data Flow Diagram (TBD)
- Sequence Diagrams (TBD)

9. Appendix C: To Be Determined List

- End-to-end encryption implementation details
- File size limits
- Third-party API selection